

The Hitchhiker Pest Program concluded on 30 June 2025. It aimed to address the risk of [hitchhiker pests](#) that can be carried via sea containers, their cargoes and associated packaging. The program adopted a systematic approach to comprehensively manage this risk through 3 focus areas:

- Expanded use of offshore controls
- Targeted onshore risk intervention and surveillance
- Partnerships with industry, government and researchers.

Why was the program needed?

The program was initiated in response to the growing threat posed by hitchhiker pests. These pests have the potential to cause significant damage to Australia's \$90 billion agricultural sector, as well as its environment and way of life.

Australia, along with several other countries, observed an increase in the global movement of hitchhiker pests associated with sea containers and their cargo. This increase was driven by a range of factors, including climate change, intensification of agricultural practices, increasing trade volumes, accelerated movement of products, supply chain complexities, and poor global shipping container hygiene.

Due to the sheer volume of sea containers entering Australia, inspecting each one upon arrival is not a feasible solution. Nor is it viable to manage the emerging risks by scaling up existing approaches or continuing along a business-as-usual trajectory.

The program instead recognised the need for a fundamental shift in how biosecurity risks are managed.

Program details

The program included 32 projects, grouped under 3 key focus areas. In addition to these focus areas, several projects enabled the overall management of the program and are classified separately below. This page provides a summary of each project, including aims, outcomes, and next steps.

Expanded use of offshore controls

Aim

This project aimed to reduce the risk of pests and contamination entering Australia by improving the design of sea containers. It focused on removing pest hiding spots and making containers easier to clean and inspect.

Better container design means fewer pests and contaminants entering Australia, faster inspections at ports and lower costs for industry.

Outcomes

- Studied current container designs and identified areas where pests tend to hide, such as under floors, in vents, and on external surfaces.
- Trialled modified containers with improved structural features, showing early signs of reduced contamination.
- Shared findings at an international symposium and contributed to global discussions on container design standards through the International Plant Protection Convention (IPPC) Sea Container Focus Group.
- Helped establish a working group of container owners, operators, and manufacturers to explore practical design changes.

Next steps

We will continue collaborating through the IPPC Sea Container Focus Group to influence global standards and maintain representation in international discussions to ensure Australia's biosecurity priorities are included.

Aim

This project aimed to prevent hitchhiker pests reaching Australia by investing in offshore biosecurity systems. It did this by collaborating with overseas National Plant Protection Organisations (NPPOs) and industry.

It established pre-border agreements and controls to reduce pest risks in shipping containers.

Outcomes

- Established and audited new Offshore Quality Systems (OQS) facilities in Papua New Guinea, with further facilities underway in New Caledonia and French Polynesia.
- Developed a draft Pacific Strategy for sea container biosecurity, with ongoing work to finalise and implement the strategy.
- Formalised a new arrangement with Mauritius, including audits and recommendations for ongoing monitoring of container hygiene.
- Contributed to the development of a Regional Standard for Phytosanitary Measures (RSPM) for sea containers through the Asia and Pacific Plant Protection Commission (APPPC).

Next steps

The next phase will focus on finalising and implementing the Pacific Strategy for sea container biosecurity. This will include further capacity building, training, and development of support materials for Pacific nations.

Ongoing audits and monitoring will continue in Mauritius and other partner countries to ensure compliance and effectiveness.

Australia will maintain active participation in international working groups and support the development of global standards for sea container hygiene. As new facilities and agreements are established, the department will work closely with regional partners to strengthen biosecurity and trade.

Learn more about [AusTreat pre-border biosecurity treatment provider scheme](#)

Aim

This project aimed to improve how offshore biosecurity treatments are managed and verified before goods arrive in Australia. The program focused on two key areas:

- AusTreat: a new system that helps regulate offshore treatment providers across multiple pests, commodities, and countries.
- Offshore Treatment Assurance Framework: a guide for managing offshore treatments consistently across different pathways.

Outcomes

- AusTreat was successfully implemented on 1 July 2024 for existing hitchhiker pathways (BMSB).
- Supported development and implementation of the treatment certificate portal.
- Managed nearly 146,000 treatment certificates over 2 seasons (2023–24 and 2024–25).
 - We identified about 1600 certificates that did not meet requirements, before goods arrived in Australia.
 - This early action meant reduced intervention at the border, saving time for an estimated 97,500 containers.

Next steps

We will continue to grow AusTreat so that it covers more pests, treatments, and high-risk products. At the same time, we will keep improving the framework and digital tools that support it, making sure they stay up to date and continue to meet Australia's biosecurity needs.

Learn more about our [existing offshore treatment provider schemes](#).

Better targeting our onshore risk intervention and surveillance

Aim

This project aimed to address the increasing threat of hitchhiker pests and contaminants on the sea container pathway by:

- Targeting onshore container inspection based on a tiered risk classification (low, medium and high) of overseas ports.

- Partnering with industry through Approved Arrangements (AAs) to inspect containers from medium and low-risk ports.
- Monitoring container inspection results and re-assessing the risk classification of overseas ports in response to emerging trends and findings.
- Strengthening training, support, and verification processes to build confidence in industry-led inspections.

Outcomes

- Developed the Sea Container Risk and Control Framework (RCF) policy outlining improved controls for managing biosecurity risks associated with sea containers
- Developed a new approach to classify overseas ports by risk level, helping to better target container inspections.
- Engaged industry through the Imported Sea Container Working Group, surveys and face-to-face meetings with stevedores, container parks and depots.
- Improved assurance in offshore treatments for managing the risk of khapra beetle in sea containers and plant products.

Next steps

The Sea Container Risk and Control Framework policy was endorsed by internal stakeholders and transitioned to relevant business areas for implementation. Implementation is dependent on enhanced ICT capabilities under the STEPS program to support efficient sea container management.

To support implementation, we will develop training and support material as well as new processes for monitoring and managing biosecurity risks arriving on the sea container pathway. We will also work closely with industry to help them adopt these changes smoothly.

We will continue to engage internationally to address hitchhiker pest risks, such as the global spread of khapra beetle through sea containers.

Aim

This project aimed to refine risk profiles to more effectively target on-arrival inspections toward sea containers most likely to carry hitchhiker pests.

We use risk profiles to set the level of intervention required for imported goods and sea containers upon arrival in Australia.

Using current data on pests, trade patterns and seasonality, the project aimed to improve profiling accuracy. This supported faster clearance of low-risk goods while strengthened biosecurity controls for higher-risk pathways.

Outcomes

- Updated 117 cargo profiles and 20 risk lists for the 2024–25 brown marmorated stink bug season, including new high-risk tariffs for emerging countries.
- Supported Compliance-Based Intervention Scheme (CBIS) expansion by developing profiles for additional commodities, ensuring interventions match actual risk.
- Maintained Entity Identifier (AEI) lists to improve targeting and support offshore assurance programs.
- Prepared new medium-risk profiles for future industry inspections, ready for rollout once Information and Communications Technology (ICT) systems are in place.
- Completed Cargo Compliance Verification (CCV) policy review with recommendations for improved sampling, governance and communications.

Next steps

As new ICT capabilities become available, we will update profiling in our routine operations, including the rollout of medium-risk profiles and ongoing seasonal updates.

We will continue to maintain AEI and CBIS profiles to ensure targeting remains current.

Our implementation of the CCV review recommendations will further improve the effectiveness and transparency of border interventions.

Learn more about the [Compliance-Based Intervention Scheme \(CBIS\)](#)

Aim

This project aimed to deliver necessary ICT enhancements and [broader application of the Compliance Based Intervention Scheme \(CBIS\)](#) across plant and animal imports, by implementing a risk-based intervention approach on additional pathways to reward compliance and improve how we regulate and manage resources at the Australian border

By focusing resources on non-compliant pathways and incentivising compliance, CBIS streamlines the border clearance process and reduces regulatory for both industry and the department.

Outcomes

- Expanded CBIS to onto 33 new plant and animal import pathways, which saves time and fees for government and industry.
- Enhanced Information and Communications Technology (ICT) systems for CBIS processing, reporting and data logging.
- Reviewed and adjusted intervention settings on a range of plant and animal import pathways to ensure intervention measures remain effective and match the level of risk they pose (not set and forget)

- Developed regular monitoring and reporting feedback loops to improve program oversight, CBIS performance and data governance.
- Improved policy and training materials to promote consistent practices for selecting pathways, provide guidance on how to process CBIS entries, and provide transparency over how the system is designed to manage CBIS entries in AIMS.

Next steps

CBIS will continue to grow, with new plant and animal import pathways and intervention measures being targeted.

We will continue to make ICT enhancements to improve how data is captured and reported and enhance how CBIS entries are processed by the import management system.

We will use data insights to track and conduct regular pathway reviews to ensure intervention settings remain fit for purpose and responsive to changing biosecurity risks and adjust rates accordingly.

Aim

This project aimed to improve early detection capabilities for high-priority hitchhiker pests at locations that receive high-risk imported goods and conveyances.

By enhancing surveillance at entry points (including [Approved Arrangements \(AAs\)](#), First Points of Entry, and near-border sites), the project strengthens our ability to prevent pests from establishing in Australia.

Outcomes

- Implemented surveillance at 69 non-AA/near-border sites across major cities, targeting exotic ants, wood-boring insects, snails, and beetles.
- Detected over 30 instances of high-priority hitchhiker pests since September 2024, highlighting the need for continued investment to manage hitchhiker pest risk.
- Tested new technologies such as suction traps, light traps, and eDNA methods to improve detection efficiency and accuracy.
- Raised awareness among importers and industry about hitchhiker pests and the importance of vigilance.
- Strengthened collaboration with state and territory plant health surveillance programs, improving planning and incident response.

Next steps

The program will further investigate and trial new surveillance technologies and refine procedures for ground-level implementation.

Our surveillance at high-risk sites will continue, with a review of detections and a proposal for ongoing management developed.

We will continue to work with federal, state, and territory partners to ensure robust legal and operational frameworks for ongoing surveillance and rapid response.

Partnering with industry, government and researchers

Aim

The Onshore [Approved Arrangement \(AA\)](#) project aimed to expand and improve the framework that allows industry participants to inspect and treat imported sea containers for hitchhiker pests.

The program enables industry to take on more responsibility for biosecurity checks. This provides capacity to increase the number of containers inspected and supports faster, more flexible responses to emerging risks.

This collaborative approach helps protect Australia's borders while streamlining operations for both government and industry.

Outcomes

- Supported the roll out of the Approved Arrangement Management Product (AAMP), giving industry participants enhanced access to their arrangement details and enabling self-service updates.
- Drafted an AA framework with new or updated AA classes and conditions for container inspections and treatments, supporting increased intervention and risk management.
- Engaged industry for feedback and consultation, including working groups and surveys to shape the future framework.
- Developed a blueprint for the future AA framework, outlining how different container types, risk levels, and activities will be managed.

Next steps

We are now further developing these digital tools under a new program called Simplified Targeting and Enhanced Processing Systems (STEPS).

STEPS will build on the work started by the Hitchhiker Program and take a broader, department-wide approach.

After enabling ICT capabilities have been developed, we will do further work to implement new and revised AA classes for container inspection and management. We will also deliver ongoing consultation and training to ensure both industry and government are equipped to manage biosecurity risks effectively.

We will continue to align AA activities with other biosecurity initiatives, supporting a more integrated and responsive system.

Aim

This project aimed to improve the administration and efficiency of [Class 19 Approved Arrangements \(AAs\)](#) by streamlining processes, enhancing assurance, and supporting increased uptake.

These improvements help ensure the department remains equipped to manage growing hitchhiker pest risks.

Outcomes

- Expanded the use of Class 19 AAs, which now handle over half of eligible entries and have saved significant staff time and costs.
- Supported the development of the Approved Arrangement Management Portal (AAMP), giving industry users easier access to update their arrangement details.
- Created training materials to help users understand seasonal pest measures and system changes.

Next steps

We will continue to explore system enhancements that allow accredited entities to view and assess goods, subject to permit conditions, further improving efficiency and compliance.

Aim

DAFF has undertaken a collection of research projects aiming to assess and validate the use of eDNA and eRNA technology for the early, non-invasive detection of hitchhiker pests in sea containers.

This approach looked to target inspections more effectively by identifying pest DNA in dust, air, or other samples, even when pests are not visible.

It investigated the technologies' ability to support rapid response and minimise disruption to trade.

Outcomes

- Developed and tested eDNA/eRNA detection technology for 5 priority hitchhiker pests, including brown marmorated stink bug, spongy moth, electric ant, red imported fire ants, and spotted lanternfly.
- Screened over 2100 shipping containers for pest contamination, providing crucial data for risk modelling and targeted management.
- Published national protocols and guidelines for eDNA testing, now accredited and available for use by laboratories across Australia and New Zealand.
- Created automated sampling devices for remote surveillance, trialled at multiple sites to collect dust for eDNA/eRNA analysis.

- Demonstrated high-throughput sequencing for non-targeted screening, identifying over 1,100 insect species in shipping containers.

Next steps

We are working to expand operational implementation where possible.

Aim

This project aimed to develop, optimise and validate a rapid, reliable molecular test for detecting key honeybee mite pests in bee swarms at the border.

By improving detection capability, the test helps protect Australia's honeybee industry from exotic pests, enabling faster responses and increasing the chance of successful eradication if an incursion occurs.

Outcomes

- Developed a new molecular diagnostic protocol for 5 priority exotic bee mite species, which can be used alongside conventional alcohol wash methods.
- Drafted a national diagnostic protocol, ready for further validation and publication.
- Reduced the need for time-consuming bee dissections, as molecular tests can quickly identify tracheal mites.
- Enhanced surveillance and response capability for exotic bee mite incursions at the border.

Next steps

We will continue engaging with CSIRO and other partners to ensure the protocol is robust and ready for operational use.

Once adopted, the molecular test could support faster, more accurate detection and strengthen Australia's biosecurity defences for honeybees.

Aim

This project aimed to improve detection of biosecurity risk material in hard-to-reach areas of containers, machinery, and vehicles. It did this by trialling a new high-tech camera 'inspection wand.'

The tool aimed to enhance inspection efficiency and reduce work health and safety risks compared to traditional methods.

Outcomes

- Improved image quality and manoeuvrability compared to the "mirror on a stick" approach.
- Demonstrated effectiveness for 5-sided container inspections, particularly for larger biosecurity risk material (e.g., mud, nests) on container tops without ladder use.

Next steps

While the project demonstrated promising results, the current cost-benefit ratio does not justify replacing existing tools with the wand.

We may explore using the livestream and record capabilities for post-inspection analysis in the future.

Aim

This project aimed to reduce manual inspection workloads and improve detection of biosecurity risk material in and around shipping containers.

We evaluated the use of machine vision AI and robotics to automate container inspections. This approach aimed to reduce manual workloads and improve the detection of biosecurity risk material in and around shipping containers.

Outcomes

- Successfully tested the robot's (Spot)'s ability to autonomously navigate around and inside containers.
- Demonstrated capability of the robot to scan container surfaces, identify biosecurity risk material, capture images, and alert officers.

Next steps

Further development is needed to refine Spot's detection accuracy and operational integration. Our future work may include enhancing AI models and exploring cost-effective deployment options.

Aim

This project aimed to develop a portable, AI-enhanced camera system for detecting exotic pests in hard-to-reach areas during inspections.

Combining hyperspectral imaging, Red, Green, and Blue (RGB) lenses, and a self-learning algorithm, the device enables rapid, accurate identification of pests like khapra beetle and snails.

Outcomes

- Developed and trialled a handheld hyperspectral and visual camera system in collaboration with Intelligent System Designs.
- Achieved high detection accuracy for adult and larval dermestid beetles and snails under controlled conditions.
- Identified performance challenges when integrating diverse environments and pest groups due to limited processing power.

Next steps

Further development of the system is needed to improve processing capability and broaden detection across multiple pest species and environments.

Any future work would focus on refining the technology for operational use in inspection and surveillance settings.

Aim

This project aimed to explore the use of automated camera technology to detect biosecurity risk material on imported containers at ports.

The goal was to improve detection efficiency, reduce manual inspection requirements, and enhance safety for officers by trialling an AI-driven system in collaboration with Australian company, Trellis Data.

Outcomes

- Trialled the BATDS system at operational ports to assess its ability to detect biosecurity risk material.
- Identified key technical and operational challenges in applying automated camera technology in real-world conditions.
- Captured valuable lessons and insights to inform future trials and technology development.

Next steps

While the current system is not ready for implementation, the knowledge gained from this project will guide future innovations in automated detection.

We would need to significantly improve detection accuracy and conduct additional trials before we can consider adopting this technology.

Aim

This project aimed to evaluate RingIR's infrared spectroscopy technology for detecting volatile organic compounds (VOCs) associated with hitchhiker pests in sea containers.

The goal was to determine whether this non-invasive method could provide rapid, accurate detection without opening containers.

Outcomes

- Successfully trialled RingIR sensors on selected container consignments to assess detection capability for pest-related VOCs.

- Demonstrated potential for early warning of contamination, reducing reliance on manual inspections.
- Identified limitations in sensitivity and specificity under operational conditions, requiring further refinement before large-scale deployment.

Next steps

Further research is needed to improve detection accuracy and integrate the technology into routine biosecurity workflows.

Aim

This project aimed to identify and validate alternative treatment options for managing snail infestations on imported goods and containers, reducing reliance on methyl bromide fumigation.

Outcomes

- Conducted trials of dry heat treatment for snail control (almost 45,000 snails were tested).
- Confirmed that dry heat treatment at 60°C for 45 minutes is effective for certain snail species.
- Developed preliminary protocols for operational use, supporting sustainability goals by reducing chemical fumigant use.

Next steps

We need to undertake further validation under varied operational conditions before this can be implemented as a biosecurity treatment.

We will continue engaging with industry to ensure the treatment options are feasible and compliant with international standards.

Aim

This project aimed to develop and assess compliance-based risk management strategies for sea containers and cargo.

The goal was to reduce biosecurity risks through targeted interventions and industry partnerships.

Outcomes

- Delivered a comprehensive risk modelling framework to inform compliance-based interventions.
- Identified key risk indicators and developed strategies for prioritising inspections based on compliance history and trade patterns.
- Provided recommendations for integrating compliance-based approaches into existing regulatory frameworks.

Next steps

The findings from this project will inform future policy development and operational guidelines, supporting a shift towards smarter, risk-based biosecurity management.

Enabling the management of hitchhiker pest risk

Aim

This project aimed to improve the collection and analysis of container movement and treatment data, enabling better risk profiling and targeted biosecurity interventions.

Outcomes

- Developed a centralised dataset integrating container movement, treatment, and inspection records.
- Identified key data gaps and inconsistencies that impact risk modelling accuracy.
- Delivered recommendations for standardising data capture and sharing across stakeholders.

Next steps

We plan to implement data quality improvements and explore automation options to enhance real-time risk assessment capabilities.

Aim

This project aimed to improve our understanding of the diversity of invertebrates and contaminants associated with the internal surfaces and underfloor of sea containers.

Outcomes

- 2147 samples collected from 2013 imported sea containers were analysed to detect and, where possible, identify animal specimens (including invertebrate pests), seeds and other contaminants.
- Over 6000 invertebrate and seeds specimens were detected - with 271 unique animal taxa and 195 unique seed taxa identified.
- The results highlight the diversity of pests and contaminants associated with the internal surfaces and underfloor of sea containers

Next steps

We will continue to communicate the importance of sea container hygiene and will work with industry to ensure containers are thoroughly cleaned prior to packing for export to Australia.

Aim

This project aimed to develop a comprehensive response strategy for managing khapra beetle incursions, ensuring rapid containment, and minimising trade disruption.

Khapra beetle is one of the world's most destructive stored product pests, and a major biosecurity threat to Australia.

A rapid, well-coordinated response strategy helps contain incursions quickly, protecting trade, agriculture and food security.

Outcomes

- Produced a national response framework outlining roles, responsibilities, and operational protocols.
- Identified critical control points and developed contingency plans for high-risk pathways.
- Delivered training materials and simulation exercises to enhance readiness across jurisdictions.

Next steps

We will continue to refine the strategy based on emerging science and international best practice.

We will also periodically engage with stakeholders and review the strategy as required.

Aim

This project aimed to strengthen the department's ability to estimate biosecurity risk, expenditure, and value across the national biosecurity system.

This modelling supports informed decision-making and prioritisation of interventions, especially for high-risk hitchhiker pest pathways.

Outcomes

- Delivered updated estimates showing how priority pests could affect Australia's environment and agriculture.
- Improved tools that track how biosecurity funding is spent, helping ensure resources go where they're needed most.
- Built a new version of the 'Value model' to show the benefits of biosecurity efforts, now hosted on department systems.
- Updated a container risk model to reflect new inspection strategies, with findings that support regular data checks to spot emerging threats.

Next steps

We will continue updating our models to reflect the latest pest risks and biosecurity spending.

We are working with other teams to make sure results can be shared and used effectively.

To keep our systems responsive and reliable, we will keep improving the 'Value model' and regularly analyse container data to spot new hitchhiker pest threats early.

Aim

This project aimed to increase awareness of hitchhiker pests among stakeholders who handle imported sea containers and goods, and to encourage reporting of suspected detections to the [See. Secure. Report. hotline](#).

Raising awareness is critical to preventing exotic pests from entering Australia and protecting our agriculture, environment, and trade. By engaging industry through multiple channels, the campaign aimed to strengthen vigilance and promote shared responsibility for biosecurity.

Outcomes

- Ran a national campaign, featuring digital advertising, merchandise distribution, conference sponsorship, and supporting activities.
- Digital advertisements generated over 29 million impressions and 33,700 clicks, driving traffic to the department's biosecurity webpage.
- Sponsored Ports Australia's 48th Biennial Conference, reaching more than 300 industry participants with campaign materials and videos.
- Delivered 2 free industry webinars attended by over 200 stakeholders, providing practical guidance on identifying and reporting pests.
- Campaign activities contributed directly to pest reporting, with 17 calls to the [See. Secure. Report. hotline](#) linked to campaign materials.

Next steps

Building on the success of the 2024 campaign, we are exploring options to repeat key elements in future years, incorporating lessons learnt to maximise reach and engagement.

Our planned improvements to key elements of the campaign would include refining digital advertising strategies, expanding merchandise offerings, and continuing webinars and stakeholder outreach.

These efforts will ensure industry remains informed, vigilant, and ready to act against hitchhiker pest threats.

Aim

This project aimed to better understand the biosecurity risks associated with containers from countries not on the department's [Country Action List](#) (CAL). By surveying these containers, the project sought to identify contamination trends and inform future risk-based inspection strategies, ensuring resources

are focused where they are most needed. The survey was undertaken in partnership with [Qube Logistics](#) in Queensland and South Australia.

Outcomes

- Conducted 1,804 container inspections at the Ports of Adelaide and Brisbane in partnership with Qube Logistics.
- Found 200 containers with evidence of biosecurity risk material, including 94 with actionable pests or contamination.
- Most common pests detected were spiders, moths, and snails; the most frequent contaminant was soil.
- Results confirmed the need for stronger monitoring of containers currently subject to minimal intervention.

Next steps

The findings from this project will inform the development of industry arrangements for inspecting medium- and low-risk containers, as outlined in the Risk and Control Framework. This will help ensure a more consistent and risk-based approach to container inspections.

Aim

This project aimed to address critical frontline resourcing needs by recruiting additional biosecurity officers dedicated to managing hitchhiker pest risks at key entry points. The objective was to strengthen operational capacity and ensure timely inspections of high-risk cargo and containers.

Outcomes

- Successfully recruited and deployed new biosecurity officers across priority ports and inspection sites.
- Enhanced onshore capability to manage increased cargo volumes and emerging pest threats.
- Improved workforce resilience and reduced inspection delays during peak periods.

Next Steps

Our ongoing workforce planning will continue to ensure sustainable staffing levels. Lessons learned from this recruitment drive will inform future strategies for rapid mobilisation during biosecurity emergencies.